

Literature Resolution of Problem 5.10

Property (B) for Pointwise Function Spaces

Status note for arXiv:1608.03883

9 August 2026

Status. `literature_already_answered`.

Original question

Krupski and Marciszewski ask in Problem 5.10 on PDF page 12 of *On the weak and pointwise topologies in function spaces II* [1]:

Characterize the property (B) of $C_p(X)$ in terms of the topology of a Tychonoff space X .

A space has property (B) when it admits a countable family of closed nowhere dense sets which absorbs every compact subset.

Explicit supporting answer

Krupski, Kucharski, and Marciszewski explicitly cite the source as “[4, Problem 5.10]”, say that they settle the question, and prove the following as Theorem 1.1 on PDF page 2 of [2]:

For every Tychonoff space X , the space $C_p(X)$ fails property (B) if and only if X has property (κ) .

Here property (κ) means that every pairwise disjoint sequence of finite subsets of X has an infinite subsequence with a point-finite open expansion. Thus the requested characterization is exact:

$$C_p(X) \text{ has property (B)} \iff X \text{ fails property } (\kappa).$$

The supporting authors recognize the identification themselves, so this is an explicit later literature answer rather than a new implication found in the present run.

Scope limitation

The result settles Problem 5.10 only. It does not settle the source paper’s Problems 1.1 and 1.2 concerning homeomorphisms between $C_p(K)$ and $C_w(L)$, or the concrete Hilbert-cube and Cech–Stone cases listed in Section 4.

References

- [1] M. Krupski and W. Marciszewski, *On the weak and pointwise topologies in function spaces II*, J. Math. Anal. Appl. 452 (2017), 646–658; arXiv:1608.03883.
- [2] M. Krupski, K. Kucharski, and W. Marciszewski, *Characterizing function spaces which have the property (B) of Banach*, arXiv:2407.18618.